

# How Much Damage Can GenAI Repair? Fact Recovery in Synthetically Damaged Historical Newspaper Text

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## Abstract

OCR errors in historical newspapers hit names, places and numbers hardest, and generative repair may replace them with fluent but wrong words. We damage one span of words around a fact at five nested levels, from one word to 75% of the sentence, in 150 BLN600 test sentences, with character noise calibrated on the corpus’s own OCR errors. With identical input, we compare fine-tuned BERT with dictionary candidates, a few-shot LLM (Qwen3-30B-A3B), a dictionary lookup and, descriptively, a fine-tuned BART denoiser, scoring repairs with BERTScore and an exact Fact Recovery Rate. We find no single level at which repair breaks down, but no main method restores more than about half of the facts: BERT with dictionary candidates recovers the anchor fact in 58% of sentences at one damaged word and 53% at 75%, the lookup alone 51%. The LLM falls from 49% to 25%, partly through format failures, and never beats the lookup on facts. BERTScore overstates repair: from 10% damage on, the denoiser scores highest on it while restoring fewer facts.

## 1 Introduction

Digitised newspapers are now a core source for historical research. Digitisation lets scholars search and mine large collections by keyword rather than reading page by page (Nguyen et al., 2021). Yet what historians search and mine is not the page itself but the text that an OCR engine read from a scan of it. OCR errors measurably change the results of quantitative analyses (Hill and Hengchen, 2019) and harm downstream NLP tasks (Strien et al., 2020), with named entities among the most affected (Hamdi et al., 2023). In BLN600, a corpus of 19th-century British newspaper excerpts, the mean character error rate (CER) is 7%, but between excerpts of a single title it ranges from 0.3% to 44.5%, and pages too faded or damaged to read had to be left out of the corpus altogether

(Booth et al., 2024). For the British Library’s own 19th-century newspaper archive, character accuracy reaches 83.6% but word accuracy falls to 78%, and accuracy for capitalised words and number groups drops further still, to 63.4% and 64.1% (Tanner et al., 2009). The words hit hardest are thus often the facts a historian searches for: names, places, dates and sums of money.

Generative AI makes it tempting to repair such text automatically, but the evidence is mixed. Llama 2, instruction-tuned on BLN600, reduced the CER of its OCR by 54.5%, against 23.3% for a fine-tuned BART (Thomas et al., 2024), while a study of 14 LLMs on several historical benchmarks found them “anything but efficient” at post-correction: they usually degraded the input, often by inventing text (Boros et al., 2024). This points to a deeper risk. A generative model always returns fluent text, even where the evidence for a word is gone (Ji et al., 2023). For an archive, a plausible but wrong name is worse than visible noise, because the reader can no longer see that anything was damaged. Similarity-based metrics may hide this error too: BERTScore, for example, rates a translation that says *Tuesday* where the reference says *Monday* as highly similar (Zhang et al., 2020).

Existing post-OCR studies report error rates over whatever noise their OCR happened to produce. Where they break results down by noise level, that level is observed rather than controlled, and facts are not scored apart from the rest of the text. They thus show that a model helps on average, but not how far repair can be trusted as damage grows, nor whether the facts inside the damaged region survive. We therefore ask:

**RQ.** When the damaged words of a sentence are known and their garbled letters are visible, how much of the original wording, and in particular its fact words such as names, places and numbers, can

a fine-tuned masked language model and a few-shot instruction-following LLM restore, and how does this change as the damaged share of the sentence grows?

We split it into three sub-questions. **SQ1** (severity): does repair quality fall smoothly as the damaged share of a sentence grows, or break down beyond some level? **SQ2** (method): given identical input, does a fine-tuned masked language model or a few-shot LLM repair better? **SQ3** (metric validity): does semantic similarity overstate repair quality compared with exact fact recovery? We fixed four hypotheses after tuning on dev but before computing any test score. **H1**: repair quality falls as the damaged share grows, and Fact Recovery Rate falls faster than BERTScore. **H2**: with identical input, the few-shot LLM restores more fact words than fine-tuned BERT, because it reads all garbled forms of a sentence together. **H2b**: GenAI repair beats a plain dictionary lookup on fact words, not only on common words. **H3**: BERTScore overstates repair quality: at every level, some repairs with a wrong fact score as high as repairs with the right one.

Answering these questions needs damage whose amount we control and whose original text we know, which real damaged pages cannot give. We therefore build a controlled experiment on BLN600 (Booth et al., 2024): in each of 150 test sentences we damage one block of words around a fact at five nested levels, from one word to 75% of the sentence, with character noise calibrated on the corpus’s own OCR errors. All methods see the same input, the intact context plus the garbled form of each damaged word, so only the repair differs. We compare masked infilling with BERT (Devlin et al., 2019), fine-tuned on in-domain gold text and guided by letter similarity and dictionary candidates in the spirit of “context beats confusion” (Evershed and Fitch, 2014), with an open-weight LLM prompted with five solved examples. As reference lines we add a dictionary lookup that uses no model at all and, descriptively, BART (Lewis et al., 2020) fine-tuned as a denoiser. We score the repaired words with BERTScore (Zhang et al., 2020) for meaning and with a Fact Recovery Rate for exact facts, and probe the LLM for memorised test text (Sainz et al., 2023).

Our contributions are (i) a fact-centred, graded damage benchmark on BLN600 with nested spans, so the same place gets worse from level to level; (ii) a controlled comparison of masked infilling

and few-shot generation under identical input and scoring, measured against a non-GenAI lookup; and (iii) a fact-level metric next to BERTScore that shows where semantic similarity and factual accuracy part ways. On the test split, the best method restores about half of the facts at every level, the few-shot LLM does not beat a dictionary lookup on facts, and BERTScore ranks fluent repairs with wrong facts too high.

## 2 Related Work

**SQ1: grading damage for repair.** We know of no study that controls damage severity for the repair step itself; graded severity has so far been used only for *reading* OCR’d text. Post-OCR research splits the task into error *detection* and error *correction* (Nguyen et al., 2021), scored separately in the ICDAR 2019 competition (Rigaud et al., 2019); we study correction only and assume the damaged span is known. Results on how well GenAI corrects disagree. An instruction-tuned Llama 2 cut CER on BLN600 further than a fine-tuned BART (Thomas et al., 2024), while 14 prompted LLMs mostly degraded historical transcripts (Boros et al., 2024); on early modern Dutch, fine-tuned seq2seq models beat fine-tuned generative ones (Debaene et al., 2025); and prompted open-weight LLMs lowered CER for English but not for Finnish (Kanerva et al., 2025). These results diverge by training regime, language and source quality, but in all of them the amount of damage is whatever the OCR produced: it can be observed, not set, so they cannot show at what point repair stops working. Controlled severity levels do exist for reading OCR text: page images degraded at two levels per degradation type and then OCR’d have been used to measure named-entity recognition (Hamdi et al., 2020), and real OCR output has been binned into quality bands for other downstream tasks (Strien et al., 2020). SQ1 applies this grading logic to the repair step.

**SQ2: restoring text at known gaps.** The closest prior task is restoring ancient text at known positions. Pythia predicts 1–10 hidden characters in Greek inscriptions (Assael et al., 2019), Ithaca extends this and pairs predictions with historians’ judgement (Assael et al., 2022), and a masked language model fills gaps in Akkadian texts, taking the number of missing signs as given and extending token predictions to multi-sign gaps with a beam search, with accuracy falling as the gap grows (Lazar et al., 2021). These studies differ

from ours in three ways. They hide characters completely, whereas OCR damage leaves letters garbled but partly readable, which our slot input exploits (Sec. 3.3); their gaps are at most a few tokens, while ours reach 75% of a sentence (SQ1); and none compares a masked language model with a prompted LLM (SQ2) or scores facts apart from overall similarity (SQ3). On the model side, BERT fills masked positions in parallel (Devlin et al., 2019), which makes multi-token blanks harder to fill (Shen et al., 2020); our search over candidate lengths (Sec. 3.4) follows the beam decoding of Lazar et al. (2021). Left-to-right infilling (Donahue et al., 2020) is the natural counterpart for a prompted LLM, and our reranking adopts the classic OCR idea of weighing context against letter evidence (Evershed and Fitch, 2014). Filling a blank from context is generation conditioned on the rest of the sentence (Donahue et al., 2020), and BERT can be sampled to produce text (Wang and Cho, 2019), so both methods count as GenAI here.<sup>1</sup>

**SQ3: meaning versus facts.** A similarity score can pass a repair that gets the facts wrong, and SQ3 tests whether that happens here. BERTScore (Zhang et al., 2020) is the standard similarity metric for generated text, but in summarisation it correlates only weakly with human judgements of faithfulness (Spearman 0.19) and factuality (0.12) (Maynez et al., 2020), and in machine translation it fails to penalise a wrong candidate that shares most content words with the reference (Hanna and Bojar, 2021), in part because BERT represents names and numbers poorly (as reviewed *ibid.*). A repaired span that restores every word but one name is exactly such a candidate. Since this evidence comes from other tasks, SQ3 checks it directly by scoring the same repairs with BERTScore and with the Fact Recovery Rate. A second concern is memorisation. Thomas et al. (2024) chose BLN600 over the ICDAR benchmarks to avoid contamination, but BLN600 itself has been public since 2024, so a newer LLM may have seen it in pretraining (Sainz et al., 2023). We probe for this with a simplified form of guided completion (Golchin and Surdeanu, 2024), prompting the model with the start of a test sentence and comparing its continuation with the true remainder.

No single element of our design is new; the

<sup>1</sup>We cite Wang and Cho (2019) only for this narrow claim; its Markov-random-field derivation was later shown to be incorrect. Either way, BERT is the in-domain baseline the few-shot LLM must beat.

| Work                       | Known loc. | Letters visible | Graded severity | Fact score | MLM vs LLM |
|----------------------------|------------|-----------------|-----------------|------------|------------|
| Post-OCR LLMs <sup>a</sup> | ×          | ✓               | ×               | ×          | ×          |
| Pythia <sup>b</sup>        | ✓          | ×               | ×               | ×          | ×          |
| Akkadian MLM <sup>c</sup>  | ✓          | ×               | ✓               | ×          | ×          |
| <b>Ours</b>                | ✓          | ✓               | ✓               | ✓          | ✓          |

Table 1: Position of this study. <sup>a</sup>Thomas et al. (2024); Boros et al. (2024); Kanerva et al. (2025) (real OCR, error rates over the corpus). <sup>b</sup>Assael et al. (2019) (1–10 hidden characters). <sup>c</sup>Lazar et al. (2021) (masked signs, results by gap length). Graded: amount of damage set by the experimenter and results reported per level.

combination is. Table 1 places our study against the three closest lines of work, and Section 3 builds each missing element: graded nested damage (Sec. 3.2), a shared slot input (Sec. 3.3), two repair families and a non-GenAI lookup under identical input (Sec. 3.4), and a fact-level score next to BERTScore (Sec. 4.2).

## 3 Method

### 3.1 Task and Data

Given a sentence whose damaged span is known and whose garbled letters are visible, the task is to predict one word per damaged position; the number of damaged words is therefore also given. All repair methods take the same input and are scored by the same code (Sec. 3.3). We build this task on BLN600 (Booth et al., 2024), 600 excerpts of London crime reporting from British newspapers of 1834–1894, each with its source image, OCR output and a gold transcription. Its gold text, aligned with real OCR, lets us calibrate our noise on real errors, and prior post-OCR work uses it (Thomas et al., 2024). We split excerpts into sentences with pysbd and pool sentences of 20–60 words with at least two *fact tokens*: a number, any word containing a digit or pound sign, or a capitalised word that is not the first word of the sentence (a short stop list excludes titles such as Mr, Mrs, The). This is a surface rule, not a named-entity tagger: it catches “Old” in “Old Bailey”, misses a lower-cased fact, and flags dates only through capitalisation or digits. The pool holds 3,123 sentences from 592 excerpts. From it we draw, with a fixed seed, 150 test and 20 dev sentences from 157 excerpts (at most three per excerpt, no excerpt in both splits, balanced across three sentence-length bands); dev is used only for tuning and is never scored as a result.

| Damage              | Type            | Value             |
|---------------------|-----------------|-------------------|
| Word drop           | probability     | 0.10              |
| Letter deletion     | probability     | 0.15 <sup>a</sup> |
| Letter substitution | probability     | 0.85              |
| confusion table     | share of subst. | 85.7%             |
| capitalization      | share of subst. | 3.6%              |
| look-alike map      | share of subst. | 4.9%              |
| random character    | share of subst. | 5.8%              |

Table 2: Damage operations. Word drop and letter deletion are sampled per word and per altered character (<sup>a</sup>deletion capped at 2 per word). Letter substitution is split among four sources by a fixed lookup order (confusion table, then capitalization, then look-alike map, then random), tried until one applies – the “share of subst.” rows are realized frequencies of that lookup, not independent sampling probabilities. The confusion table is calibrated on 29 held-out BLN600 excerpts.

### 3.2 Damage Simulation

For each sentence we draw one anchor among its fact tokens – always damaged, never dropped – and damage one contiguous span of  $k$  words that centers on it, with  $k$  set so the span covers 1 word, then 10%, 25%, 50% or 75% of the sentence (half-up rounding). Every larger span centers on the same anchor and fully contains every smaller one (Figure 1), so a higher damage level adds more damaged words around a fixed center; it does not make any single word more damaged; the per-word alteration intensity below is drawn the same way regardless of the span size. Each damaged word may be dropped outright (except the anchor). A surviving word has 20–30% (at least one) of its letters and digits altered, punctuation untouched, split between deletion and substitution as in Table 2 (the substitution source is a fixed lookup order – confusion table, then capitalization, then a look-alike map, then a random character – not a random draw between them). The 20–30% band was chosen to sit around the median per-word character error rate we measured on 29 held-out BLN600 excerpts (0.25, IQR 0.17–0.40). The confusion table (85.7% of substitutions) and this band come from real OCR errors on aligned words; the drop rate and fallbacks are our choice.

Table 3 shows the result on the test split: sentence-level CER grows roughly in step with real BLN600 OCR error rates, but the share of damaged words that are themselves facts falls from 100% at the 1-word level to 17% at 75%. About one in ten fact slots is dropped outright, leaving no visible letters for that fact at all.

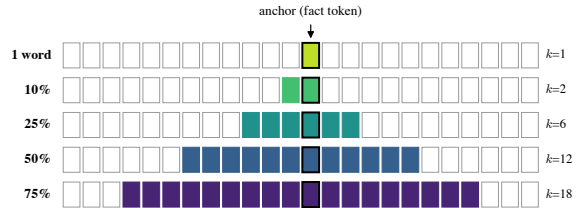


Figure 1: The damaged span at each level, for one example 24-word sentence. Every span centers on the same anchor word (the seeded fact token) and fully contains every smaller span, so a higher level does not change how heavily any single word is damaged – only how many words around the anchor are included.

### 3.3 Shared Slot Input

All methods see the same *slot view*: the sentence’s undamaged context is left untouched, and each damaged gold word is replaced by a slot showing its garbled form (a dropped word becomes an empty slot, written  $\langle ? \rangle$ ). This view is built from the gold tokens and the per-word edit operations, never from the raw corrupted text, so it cannot leak an easier version of the task. A predicted word is spliced back inside the original word’s edge punctuation, so context words are never rewritten and every difference in score comes from what a method puts in the slots. For example, the damaged sentence “A POWERFULLY-BUILT man,  $\langle \text{same} \rangle$   $\langle \text{Bbonias} \rangle$   $\langle \text{Huicl} \rangle$   $\langle \text{e} \rangle$  commission agent, . . .” has the gold continuation “named Thomas Hurd, a”; the garbled forms are visible to both repair methods, but the gold words are not. Showing garbled rather than blanked letters keeps this an OCR-repair task rather than blind text infilling. Treating the damaged region as known follows the usual split of post-OCR work into detection and correction (Nguyen et al., 2021; Rigaud et al., 2019) and the known-position setting of text restoration (Assael et al., 2019; Lazar et al., 2021); it isolates repair from detection, which we leave to future work.

### 3.4 Repair Methods

(a) **BERT + reranking.** We fine-tune bert-base-cased (Devlin et al., 2019) (110M parameters; cased because fact tokens are partly defined by capitalisation) on clean gold text from the 443 excerpts outside the sample (2,939 sentences, of which those from 10% of the excerpts are held out for validation, with zero excerpt or sentence overlap with test or dev, which we verified directly). Training mirrors inference: for each target word, span words before it are left



as gold text, the target is masked with its true number of subword pieces, and later span words each become one [MASK]. Fine-tuning settings are in Table 4; the best checkpoint (epoch 2) lowered the per-piece validation loss from 5.61 (pretrained) to 4.14, and training loss kept falling while validation loss rose again over the next two epochs, confirming the early stop. At inference, slots are filled left to right, following the beam decoding of Lazar et al. (2021) with the settings in Table 4. A slot with a visible internal hyphen or apostrophe is split into sub-slots filled the same way; only 0.9% of slots (3.5% of fact slots) still need more than three pieces. Each candidate is scored as its mean log-probability per piece plus  $\lambda(1 - d)$ , where  $d$  is the normalised Levenshtein distance to the garbled form, so a fluent completion is pulled back toward what the damaged letters actually show. This is a direct, modern instance of “context beats confusion” (Evershed and Fitch, 2014). We chose  $\lambda = 8$  from a grid over  $\{1, 2, 4, 8, 16\}$  on dev, where it gave the best exact-word rate at a point where Fact Recovery Rate had already plateaued. Dropped slots, which have no garbled letters to match against, are scored on the language-model term alone. We keep this variant without list words as an ablation (*BERT, no dict.*).

**(b) BERT + dictionary.** Reranking cannot choose a word that BERT never proposes: on dev, BERT’s top candidates for *Wright* (gold *Wright*) were *Smith*, *White* and *andooll*. Our main MLM method therefore adds, for each slot, the five entries of a word list that are closest in spelling to the garbled form to BERT’s candidates, and scores them with the same formula. The list holds the 16,469 distinct words of the gold text of the 443 excerpts outside the sample, so no dev or test text enters it. On a dev grid over list size (5, 10, 20, 50) and  $\lambda$  (1 to 128) we chose 5 entries and  $\lambda = 64$  by the same rule as above. At this weight the letters decide most slots, and BERT’s context breaks near ties and fills dropped slots.

**(c) LLM few-shot.** We prompt Qwen3-30B-A3B-Instruct-2507 (Yang et al., 2025), a mixture-of-experts model with 30B parameters of which 3B are active per token, served through GWDG’s SAIA API. The prompt holds a system message, a task instruction, five solved dev examples (one per level, held out from all scoring), and the same slot view as BERT with numbered slots, and asks for a JSON list of exactly  $k$  strings at temperature 0. We first used Llama-3.1-8B-Instruct (Grattafiori

et al., 2024), but its dev format-failure rates (21% and 27% for the two prompt variants) were above the 15% limit we had set in advance for switching model. We use only open-weight models on academic infrastructure, so that the licensed BLN600 text (CC BY-NC-ND) is never sent to a commercial third-party service. The system message was necessary: without it, an early dev check found the model answering with Python code in 15 of 15 cases, in line with the finding that stating the input type and output format helps LLM post-correction (Boros et al., 2024). An answer of the wrong length, not valid JSON, or cut off by the token limit counts as a *format failure* and is scored as if no repair had been made, with no retry and no manual correction. Because output formatting alone can swing LLM post-correction results (Kanerva et al., 2025), we also report the LLM’s scores over valid answers only, which separates failing to repair from failing to follow the format. A secondary variant supplies the whole surrounding article as context instead of just the sentence; its other sentences are gold, not damaged, so it is an upper bound rather than a fair comparison with BERT, and we compare it only against the LLM’s own few-shot run. Zero-shot prompting was tried on dev and dropped, so this design cannot isolate how much of the LLM’s performance comes from the five examples rather than the instruction alone. The two methods also differ in supervision by design: BERT is fine-tuned on in-domain text, the LLM sees five examples. SQ2 therefore compares two practical repair set-ups, not two architectures under equal training.

**(d) Dictionary lookup.** To ask whether GenAI beats a spell checker, each garbled word is replaced by the closest entry of the same word list (Levenshtein distance, ties by frequency); a dropped slot stays empty. The lookup uses letters only and no context, the pure “confusion” side of Evershed and Fitch (2014). Its limit is coverage: on dev, the list holds 95% of damaged words but only 77% of damaged fact words.

**(e) BART denoiser (descriptive).** bart-base (140M) (Lewis et al., 2020), the model family that Thomas et al. (2024) compared on BLN600, reads the slot view and writes the sentence back with one word per slot. We fine-tune it on the BERT training sentences, damaged by our own generator at all five levels with new damage in each epoch (best epoch 8 of 10). Unlike the other methods, BART is thus trained on exactly the noise process of the test data, so its scores are optimistic. It was added after the

| Level                     | 1w  | 10% | 25% | 50%  | 75%  |
|---------------------------|-----|-----|-----|------|------|
| Sentence CER (%)          | 1.0 | 3.2 | 7.9 | 15.5 | 23.1 |
| Facts in span (mean)      | 1.0 | 1.6 | 2.2 | 3.2  | 4.2  |
| Facts / damaged words (%) | 100 | 48  | 26  | 19   | 17   |

Table 3: Damage by level (test split). CER is over the whole sentence against gold; real BLN600 OCR is about 7% (Booth et al., 2024).

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| <i>BERT (both variants)</i> |                                                                                                 |
| Base model                  | bert-base-cased (110M), clean gold text                                                         |
| Fine-tuning                 | lr 5e-5, batch 16, early stop (patience 2, max 10 ep.), best epoch 2                            |
| Reranking                   | 1-3 masks/slot, beam 5, top 10; $\lambda = 8$ (no dict.), $\lambda = 64 + 5$ list words (dict.) |
| <i>LLM few-shot</i>         |                                                                                                 |
| Model                       | Qwen3-30B-A3B-Instruct-2507 (SAIA)                                                              |
| Prompt                      | system msg + 5 dev examples, numbered slots, JSON                                               |
| Decoding                    | temperature 0, max 1024 tokens                                                                  |
| <i>BART and lookup</i>      |                                                                                                 |
| BART                        | bart-base (140M), same optimiser as BERT, trained on our damage; best epoch 8; greedy           |
| Lookup                      | 16,469-word list, Levenshtein, no context                                                       |

Table 4: Implementation settings.  $\lambda$  and list size chosen on dev only.

hypotheses were set, and we report it descriptively only.

**(f) No repair.** The damaged text itself is scored as if it were a method, giving the floor every repair has to beat. Table 4 summarises the settings.

## 4 Empirical Investigation

### 4.1 Experimental Design

The design crosses three factors inside the same 150 test sentences: five damage levels (1 word, then 10, 25, 50 and 75% of the sentence), three length bands (20-29, 30-44 and 45-60 words), and the repair methods of Sec. 3.4 (Table 5). A clean level 0 serves only the contamination probe (Sec. 4.2). All damage levels reuse the same sentence and the same anchor word, so every comparison across levels is paired within the same sentence. The 20 dev sentences, disjoint from test, were used only to pick settings (Sec. 3.4); five of them are the LLM’s examples, so 15 are scored. Each method’s setting was frozen by a code gate before its test rows were touched, and all test scores were computed once, by one notebook, after every method’s predictions were stored.

| Factor      | Levels                                                                                                                                                                                     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Severity    | 1 word, 10, 25, 50, 75% of words; clean level 0 for the contamination probe only                                                                                                           |
| Length band | 20-29, 30-44, 45-60 words                                                                                                                                                                  |
| Methods     | BERT + dict.; LLM few-shot; dictionary lookup; no repair. Secondary: LLM + article, BERT (no dict.), BART (descriptive)                                                                    |
| Data        | 150 test sentences, 750 rows per method; dev 20 (15 scored, tuning only)                                                                                                                   |
| Metrics     | BERTScore, Fact Recovery Rate (primary); anchor recovery, exact match, pooled span CER, repair gain, format failures, LLM valid-only scores, sentence BERTScore, contamination (secondary) |

Table 5: Experimental design.

**Statistics.** Slots of one sentence share their context, so the sentence, not the slot, is our unit. Every cell gets a 95% bootstrap interval over sentences (10,000 resamples, the same resamples for all methods; Koehn, 2004; Dror et al., 2018). We test only the two comparisons that a hypothesis named before the test run: BERT + dict. vs LLM (H2) and LLM vs lookup (H2b). Per level we use an exact McNemar test on anchor recovery and a Wilcoxon signed-rank test on span BERTScore, the standard non-parametric tests for paired binary outcomes and paired scores (Dror et al., 2018), with Holm’s correction (Holm, 1979) over the ten tests of each family. All other comparisons are descriptive.

### 4.2 Metrics

We score the repaired span, not the whole sentence, since only the span was touched. BERTScore F1 (Zhang et al., 2020) compares the repaired span with the gold span using roberta-large (layer 17), rescaled against a baseline so that a score can go below 0; this is our primary meaning-level metric. We also report sentence-level BERTScore, where the intact context can raise the score. Fact Recovery Rate is our primary fact-level metric: the share of gold fact tokens in the span that the repair restores exactly in their slot, punctuation ignored, averaged over sentences. Because the share of facts among damaged words falls as the span grows (Table 3), we also report recovery of the anchor alone, which is the same word at every level and so gives a clean paired series for SQ1, and we split Fact Recovery Rate into fact slots with visible letters and dropped slots, which the methods must fill from context alone.

As secondary metrics we report exact-word match (the share of span words restored exactly,

averaged over sentences), span CER pooled per level (all character edits divided by all gold characters of that level), repair gain (pooled CER before minus after repair),<sup>2</sup> the LLM’s format-failure rate and its scores over valid answers only (Sec. 3.4), and a contamination probe. The probe is a simplified guided completion (Golchin and Surdeanu, 2024; Sainz et al., 2023): the LLM receives the first half of a clean test sentence and is asked to continue it, and a continuation whose normalised Levenshtein similarity to the true remainder (cut to the same length) is at least 0.9 counts as near-verbatim. Unlike Golchin and Surdeanu (2024), the prompt does not name BLN600 and there is no second condition to compare with, so a formulaic phrase can pass the threshold by chance; we therefore read a low rate as reassuring and inspect any hit by hand. Because of limited API resources, the probe covers 116 of the 150 test sentences.

BERTScore and Fact Recovery Rate are reported side by side because they can disagree by design (SQ3): a repair that substitutes a wrong but plausible name can still score well on meaning while scoring zero on Fact Recovery Rate. To put one number on this gap (H3) without a threshold, we take for each method and level the chance that a repair with a wrong anchor gets at least as high a span BERTScore as a repair with the right anchor (ties count half), which is  $1 - \text{AUC}$ . A value of 0 means BERTScore always ranks the right fact higher; 0.5 means it cannot tell right from wrong facts.

### 4.3 Results

**Severity (SQ1).** Every method stays well above the no-repair floor, and we find no single level at which repair breaks down (Table 6, Figure 2). Span BERTScore falls for all methods as the span grows, but the fact scores of the letter-based methods fall only slightly: BERT + dictionary restores the anchor in 58% of sentences at one damaged word and in 53% at 75%, and the context-free lookup in 51% at every level. The LLM falls from 49% to 25%, most between 25% and 50%, BART from 47% to 36%; Fact Recovery Rate follows the same pattern. Dropped fact words are almost never restored (at most 6% for BERT + dict., 23% for BART, on few slots). The LLM’s decline is steepest in the longest

sentences (span BERTScore below zero at 75%), while BERT + dictionary stays at 0.39 to 0.43 in all length bands.

**Method (SQ2).** The LLM never restores more anchors than BERT + dictionary, which is ahead at every level (26 to 55 sentences where only it restores the anchor, against 13 to 16), significantly at 50% and 75% on both tests (Holm-adjusted  $p < 0.01$ ) but not below ( $p > 0.13$ ). The lookup also beats the LLM on anchors and span BERTScore at 50% and 75% ( $p \leq 0.04$ ); the LLM’s only significant advantage over it is span BERTScore at 10% ( $p = 0.039$ ), with no significant difference in anchor recovery. Descriptively, BERT + dictionary leads the lookup by only 0.01 to 0.07 in Fact Recovery Rate.

**Format failures.** The LLM failed the format in 106 of 750 answers (14%, all but 3 by a wrong word count), 103 of them at 50% and 75% (Table 7). Over its valid answers, its span BERTScore at 75% is 0.40, as for BERT + dictionary, but its Fact Recovery Rate stays lower (0.39 against 0.48); valid answers are a selected subset, so this is descriptive only.

**Metric validity (SQ3).** Beyond one word, the chance that a repair with the wrong anchor scores at least as high in span BERTScore as one with the right anchor is 0.11 to 0.45 for every main method, highest for BERT + dictionary and the lookup at 75% (0.45, 0.43). BART shows the gap directly: from 10% on it has the highest span BERTScore of the main methods (0.67 to 0.50) but restores fewer facts than BERT + dictionary at every level, writing for instance *Alexander Fleming* for *Alexander Tcmpl* (gold *Temple*) and *Burnley* for *Hacley* (gold *Hanley*).

**Ablations and contamination.** Without dictionary candidates, BERT restores almost no facts beyond one word (Fact Recovery Rate 0.13 to 0.04) and makes the span worse than the damage (pooled repair gain  $-0.18$  to  $-0.34$ ). The article as context raises the LLM’s Fact Recovery Rate by 0.1 to 0.2, the highest fact scores of all up to 25%, but its context is clean gold text, which real pages would not give. The probe found no near-verbatim continuation in 116 test sentences (median similarity 0.26, maximum 0.51).

<sup>2</sup>We pool CER per level rather than averaging it per row, because on a one-word span a single wrong or over-long word can already give a row CER of 1 or more, so short spans would dominate the mean.

| Level                     | BERT+dict.             | Main methods (95% CI) |                 |                        | Floor<br>None | Ablations |          |
|---------------------------|------------------------|-----------------------|-----------------|------------------------|---------------|-----------|----------|
|                           |                        | LLM                   | Lookup          | BART                   |               | BERT      | LLM+art. |
| <i>BERTScore (span)</i>   |                        |                       |                 |                        |               |           |          |
| 1w                        | <b>0.77</b> [.71, .82] | 0.74 [.69, .78]       | 0.72 [.66, .77] | 0.70 [.64, .75]        | 0.30          | 0.48      | 0.83     |
| 10%                       | 0.62 [.57, .67]        | 0.60 [.55, .65]       | 0.52 [.47, .56] | <b>0.67</b> [.62, .71] | 0.01          | 0.36      | 0.69     |
| 25%                       | 0.47 [.43, .51]        | 0.46 [.43, .49]       | 0.41 [.38, .45] | <b>0.62</b> [.58, .67] | −0.16         | 0.16      | 0.56     |
| 50%                       | 0.42 [.39, .45]        | 0.24 [.18, .30]       | 0.37 [.34, .40] | <b>0.56</b> [.50, .61] | −0.26         | 0.02      | 0.40     |
| 75%                       | 0.40 [.37, .43]        | 0.07 [.00, .13]       | 0.37 [.34, .40] | <b>0.50</b> [.44, .56] | −0.31         | −0.02     | 0.13     |
| <i>Fact Recovery Rate</i> |                        |                       |                 |                        |               |           |          |
| 1w                        | <b>0.58</b> [.50, .66] | 0.49 [.41, .57]       | 0.51 [.43, .59] | 0.47 [.39, .55]        | 0.00          | 0.25      | 0.69     |
| 10%                       | <b>0.54</b> [.47, .61] | 0.46 [.39, .54]       | 0.49 [.42, .56] | 0.45 [.38, .51]        | 0.00          | 0.13      | 0.58     |
| 25%                       | <b>0.49</b> [.43, .56] | 0.38 [.32, .44]       | 0.46 [.40, .52] | 0.42 [.36, .48]        | 0.00          | 0.08      | 0.57     |
| 50%                       | <b>0.48</b> [.42, .53] | 0.32 [.26, .38]       | 0.47 [.42, .53] | 0.42 [.36, .48]        | 0.00          | 0.05      | 0.46     |
| 75%                       | <b>0.48</b> [.43, .53] | 0.22 [.17, .27]       | 0.43 [.39, .48] | 0.40 [.34, .45]        | 0.00          | 0.04      | 0.32     |

Table 6: Test results, 150 sentences per level (95% bootstrap intervals over sentences). Best main method per row in bold. LLM = few-shot; a format failure counts as no repair (Table 7); anchor recovery is in Figure 2. BART is descriptive only; BERT without dictionary and LLM with the article are ablations.

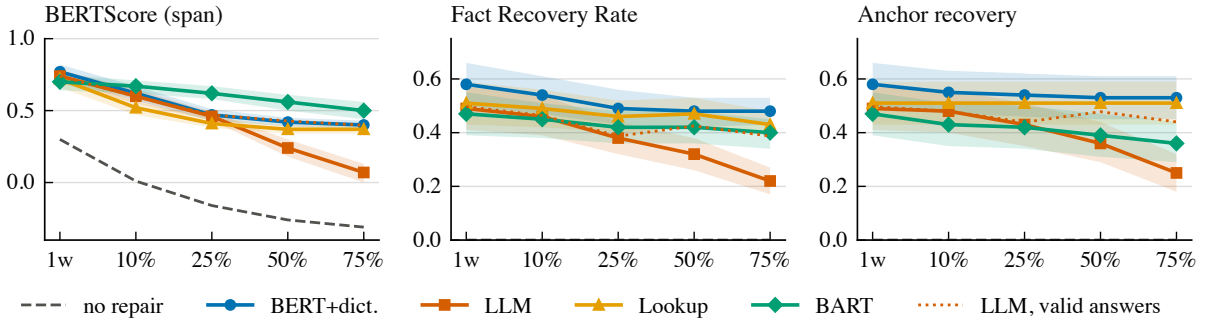


Figure 2: Test scores by damage level (150 sentences per level), with 95% bootstrap bands. The dotted line is the LLM over its valid answers only (a shrinking subset at 50% and 75%, Table 7).

| Level | Format failures |       |      | LLM, valid answers |      |          |
|-------|-----------------|-------|------|--------------------|------|----------|
|       | LLM             | +art. | BART | BERTSc.            | FRR  | <i>n</i> |
| 1w    | 1               | 0     | 0    | 0.74               | 0.50 | 149      |
| 10%   | 0               | 0     | 0    | 0.60               | 0.46 | 150      |
| 25%   | 2               | 2     | 6    | 0.47               | 0.39 | 148      |
| 50%   | 37              | 23    | 13   | 0.43               | 0.42 | 113      |
| 75%   | 66              | 68    | 21   | 0.40               | 0.39 | 84       |

Table 7: Format failures per level (of 150; all levels: 106, 93, 40) and the LLM over its valid answers (*n* rows; FRR = Fact Recovery Rate).

## 5 Interpretation and Conclusion

The objective was to find how far text repair can be trusted as the damaged share of a sentence grows. For the letter-based methods the answer is a ceiling rather than a breakpoint: they are about as reliable at 75% damage as at one damaged word, but at no level better than about one fact in two (anchor 58% to 53% for BERT + dictionary). The LLM loses facts as the damage grows (49% to 25%), most sharply between 25% and 50%, where its format failures begin. The objective is reached only in

part. **H1** is partly supported (descriptively): quality falls with damage, but not faster in Fact Recovery Rate than in BERTScore (BERT + dictionary: Fact Recovery Rate 0.58 to 0.48, exact words 0.58 to 0.57, span BERTScore 0.77 to 0.40). **H2** is not supported for the tested pair: the LLM never restores more facts than BERT + dictionary, and significantly fewer at 50% and 75%; only BERT without dictionary candidates restores fewer facts than the LLM. **H2b** is not supported for the tested pair: the few-shot LLM does not beat the lookup on facts at any level and loses to it at 50% and 75%. **H3** is supported beyond one word, where BERTScore ranks a wrong-fact repair at least as high as a right-fact repair in 11% to 45% of pairs; at one word the span is the fact itself, so it cannot hold there.

Format failures explain much of the LLM’s decline at high damage (44% of its answers at 75%), but on facts even its valid answers stay behind. Two differences can explain this. First, the methods use the garbled letters differently: BERT sees them only through its candidates and reranking



term, the lookup uses them alone, and the LLM and BART read all garbled forms at once. We expected this joint view to favour the LLM as spans grow; it did not. Instead, the methods that rank candidates by their letters lead: BERT with dictionary candidates (and its higher letter weight) restores 0.33 to 0.44 more of the facts than BERT without them, but only 0.01 to 0.07 more than the lookup alone. For a fact word with only some letters altered, the closest spelling in a word list is a strong guess that a generative model does not reliably beat. Second, supervision and scale differ: BERT is fine-tuned on in-domain text and BART on our own damage, while the LLM has about 270 times more parameters than BERT (27 times more active per token) but sees five examples. A related post-OCR comparison likewise found fine-tuned sequence-to-sequence models to correct more errors than generative LLMs, even fine-tuned ones (Debaene et al., 2025). The probe found no near-verbatim continuation in 116 test sentences, which does not exclude other forms of memorisation.

Semantic similarity overstates how much factual content a repair restores. BART, trained on the test noise process itself, scores highest among the main methods on span BERTScore from 10% damage on while restoring fewer facts than BERT + dictionary, with repairs such as *Alexander Fleming* for *Alexander Temple*: fluent, plausible and wrong. This matches findings from summarisation and translation (Maynez et al., 2020; Hanna and Bojar, 2021) and carries over to text repair, even with BERTScore computed on the repaired span only.

In practice, a restored name, place or number should not be trusted without a check at any damage level. Our results favour a letter-first design: propose repairs from the garbled letters, with context breaking ties as in BERT + dictionary, and show them as marked suggestions with ranked alternatives, so the historian decides, as Ithaca pairs model predictions with historians’ judgement (Assael et al., 2022). A generative model should not rewrite facts unchecked: its fluent errors hide that anything was damaged. How repair performs on real damaged pages, with heavier noise and no known damaged region, remains untested.

In short, when the damaged words are known and their letters visible, the best repair restores about half of the fact words at every damage level; the letters carry most of this, and neither a few-shot LLM nor a fine-tuned denoiser restores more facts

than a dictionary lookup guided by BERT. Three extensions follow from our assumptions: combining repair with detection of the damaged region; testing on real OCR errors from damaged pages; and repairing from the scanned image itself, through image-level degradation as a test bed (Hamdi et al., 2020) or vision-capable models that read the damaged page directly (cf. Kanerva et al., 2025).

## Limitations

*Damage.* Our damage is synthetic character noise in one contiguous span, not a model of torn, stained or faded pages: intensity is drawn per word rather than spatially correlated as in a real torn patch, and the confusion table comes from mild OCR errors on aligned words, so it shows which characters get confused but likely understates how heavy real damage is. Synthetic and natural noise can differ enough that results on one do not carry over to the other (Belinkov and Bisk, 2017). *Task.* The damaged region and the number of damaged words are given, so no detection step is tested. Damage is centred on a fact by design, so facts make up 100% of damaged words at the 1-word level but only 17% at 75%, and what else falls in the span, a stock phrase or a rare name, is not controlled. *Measurement.* Fact tokens come from a surface rule (a hand check of 50 random test fact tokens: 42 true facts, 6 not, 2 unclear) and are scored by exact match, so a correct spelling variant (*M’Donald* for *McDonald*) counts as a failure. With 150 test sentences, only 26 in the longest band, and one noise draw per sentence, we report trends rather than powered effects. *Comparison.* The methods differ beyond the repair step: the LLM and BART read the garbled letters directly, BERT only through its candidates and reranking score; BERT predicts a word’s pieces in parallel (Shen et al., 2020) and cannot produce a word part of more than three pieces; supervision, scale and pretraining differ; and the LLM may have seen BLN600, which has been public since 2024. Format failures count as no repair, a constraint only the LLM and BART can fail, and the article variant uses gold context and is therefore an upper bound. BART is trained on the same damage generator as the test data, so its scores are optimistic, and each fine-tuned model comes from one training run with one seed. The contamination probe covers only 116 of the 150 test sentences. All data are English London crime reports, mostly from two weeklies (572 of 600 excerpts), and we test one LLM.

## References

- Yannis Assael, Thea Sommerschild, and Jonathan Prag. 2019. [Restoring ancient text using deep learning: a case study on Greek epigraphy](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6368–6375, Hong Kong, China. Association for Computational Linguistics.
- Yannis Assael, Thea Sommerschild, Brendan Shillingford, Mahyar Bordbar, John Pavlopoulos, Marita Chatzipanagiotou, Ion Androutsopoulos, Jonathan Prag, and Nando de Freitas. 2022. [Restoring and attributing ancient texts using deep neural networks](#). *Nature*, 603(7900):280–283.
- Yonatan Belinkov and Yonatan Bisk. 2017. Synthetic and natural noise both break neural machine translation. *arXiv preprint arXiv:1711.02173*.
- Callum William Booth, Alan Thomas, and Robert Gaizauskas. 2024. [BLN600: A Parallel Corpus of Machine/Human Transcribed Nineteenth Century Newspaper Texts](#). In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 2440–2446, Torino, Italia. ELRA and ICCL.
- Emanuela Boros, Maud Ehrmann, Matteo Romanello, Sven Najem-Meyer, and Frédéric Kaplan. 2024. [Post-Correction of Historical Text Transcripts with Large Language Models: An Exploratory Study](#). In *Proceedings of the 8th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL 2024)*, pages 133–159, St. Julians, Malta. Association for Computational Linguistics.
- Florian Debaene, Aaron Maladry, Els Lefever, and Veronique Hoste. 2025. [Evaluating Transformers for OCR Post-Correction in Early Modern Dutch Theatre](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 10367–10374, Abu Dhabi, UAE. Association for Computational Linguistics.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Chris Donahue, Mina Lee, and Percy Liang. 2020. [Enabling Language Models to Fill in the Blanks](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2492–2501, Online. Association for Computational Linguistics.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. [The Hitchhiker’s Guide to Testing Statistical Significance in Natural Language Processing](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia. Association for Computational Linguistics.
- John Evershed and Kent Fitch. 2014. [Correcting noisy OCR: context beats confusion](#). In *Proceedings of the First International Conference on Digital Access to Textual Cultural Heritage, DATeCH ’14*, pages 45–51, New York, NY, USA. Association for Computing Machinery.
- Shahriar Golchin and Mihai Surdeanu. 2024. [Time Travel in LLMs: Tracing Data Contamination in Large Language Models](#). In *International Conference on Learning Representations*, volume 2024, pages 43008–43029.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, and others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- Ahmed Hamdi, Axel Jean-Caurant, Nicolas Sidère, Mickaël Coustaty, and Antoine Doucet. 2020. Assessing and Minimizing the Impact of OCR Quality on Named Entity Recognition. In *Digital Libraries for Open Knowledge*, pages 87–101, Cham. Springer International Publishing.
- Ahmed Hamdi, Elyys Linhares Pontes, Nicolas Sidère, Mickaël Coustaty, and Antoine Doucet. 2023. In-depth analysis of the impact of OCR errors on named entity recognition and linking. *Natural Language Engineering*, 29(2):425–448.
- Michael Hanna and Ondřej Bojar. 2021. [A Fine-Grained Analysis of BERTScore](#). In *Proceedings of the Sixth Conference on Machine Translation*, pages 507–517, Online. Association for Computational Linguistics.
- Mark J Hill and Simon Hengchen. 2019. Quantifying the impact of dirty OCR on historical text analysis: Eighteenth Century Collections Online as a case study. *Digital Scholarship in the Humanities*, 34(4):825–843.
- Sture Holm. 1979. [A Simple Sequentially Rejective Multiple Test Procedure](#). *Scandinavian Journal of Statistics*, 6(2):65–70.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. [Survey of Hallucination in Natural Language Generation](#). *ACM Comput. Surv.*, 55(12).
- Jenna Kanerva, Cassandra Ledins, Siiri Käpyaho, and Filip Ginter. 2025. [OCR Error Post-Correction with LLMs in Historical Documents: No Free Lunches](#). In *Proceedings of the Third Workshop on Resources and*

- Representations for Under-Resourced Languages and Domains (RESOURCEFUL-2025)*, pages 38–47, Tallinn, Estonia. University of Tartu Library, Estonia.
- Philipp Koehn. 2004. [Statistical Significance Tests for Machine Translation Evaluation](#). In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395, Barcelona, Spain. Association for Computational Linguistics.
- Koren Lazar, Benny Saret, Asaf Yehudai, Wayne Horowitz, Nathan Wasserman, and Gabriel Stanovsky. 2021. [Filling the Gaps in Ancient Akkadian Texts: A Masked Language Modelling Approach](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 4682–4691, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. [BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, Online. Association for Computational Linguistics.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. [On Faithfulness and Factuality in Abstractive Summarization](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919, Online. Association for Computational Linguistics.
- Thi Tuyet Hai Nguyen, Adam Jatowt, Mickael Coustaty, and Antoine Doucet. 2021. [Survey of Post-OCR Processing Approaches](#). *ACM Comput. Surv.*, 54(6).
- Christophe Rigaud, Antoine Doucet, Mickaël Coustaty, and Jean-Philippe Moreux. 2019. [ICDAR 2019 Competition on Post-OCR Text Correction](#). In *2019 International Conference on Document Analysis and Recognition (ICDAR)*, pages 1588–1593.
- Oscar Sainz, Jon Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. [NLP Evaluation in trouble: On the Need to Measure LLM Data Contamination for each Benchmark](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10776–10787, Singapore. Association for Computational Linguistics.
- Tianxiao Shen, Victor Quach, Regina Barzilay, and Tommi Jaakkola. 2020. [Blank Language Models](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5186–5198, Online. Association for Computational Linguistics.
- Daniel Strien, Kaspar Beelen, Mariona Coll Ardanuy, Kasra Hosseini, Barbara McGillivray, and Giovanni Colavizza. 2020. [Assessing the Impact of OCR Quality on Downstream NLP Tasks](#).
- Simon Tanner, Trevor Muñoz, and Pich Hemy Ros. 2009. Measuring mass text digitization quality and usefulness. *D-lib Magazine*, 15(7/8):1082–9873.
- Alan Thomas, Robert Gaizauskas, and Haiping Lu. 2024. [Leveraging LLMs for Post-OCR Correction of Historical Newspapers](#). In *Proceedings of the Third Workshop on Language Technologies for Historical and Ancient Languages (LT4HALA) @ LREC-COLING-2024*, pages 116–121, Torino, Italia. ELRA and ICCL.
- Alex Wang and Kyunghyun Cho. 2019. [BERT has a Mouth, and It Must Speak: BERT as a Markov Random Field Language Model](#). In *Proceedings of the Workshop on Methods for Optimizing and Evaluating Neural Language Generation*, pages 30–36, Minneapolis, Minnesota. Association for Computational Linguistics.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. [Qwen3 Technical Report](#).
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. [BERTScore: Evaluating Text Generation with BERT](#).